

Math Sec 1

1. The mean and variance of 10 observations are 9 and 34.2, respectively. If 8 of these observations are 2, 3, 5, 10, 11, 13, 15, 21, then the mean deviation about the median of all the 10 observations is
(A) 5 (B) 6 (C) 4 (D) 7
2. Let $S = \{x^3 + ax^2 + bx + c : a, b, c \in \mathbb{N}, a, b, c \leq 20\}$ be a set of polynomials. Then the number of polynomials in S which are divisible by $x^2 + 2$ is
(A) 10 (B) 120 (C) 6 (D) 20
3. If $g(x) = 3x^2 + 2x - 3$, $f(0) = -3$ and $4g(f(x)) = 3x^2 - 32x + 72$, then $f(g(2))$ is equal to:
(A) $-\frac{25}{6}$ (B) $\frac{7}{2}$
(C) $-\frac{7}{2}$ (D) $\frac{25}{6}$
4. Let z be a complex number such that $|z - 6| = 5$ and $|z + 2 - 6i| = 5$. Then the value of $z^3 + 3z^2 - 15z + 141$ is equal to
(A) 37 (B) 42 (C) 50 (D) 61
5. If α, β , where $\alpha < \beta$, are the roots of the equation $\lambda x^2 - (\lambda + 3)x + 3 = 0$ such that $\frac{1}{\alpha} - \frac{1}{\beta} = \frac{1}{3}$, then the sum of all possible values of λ is
(A) 4 (B) 2 (C) 6 (D) 8
6. A bag contains 10 balls out of which k are red and $(10 - k)$ are black, where $0 \leq k \leq 10$. If three balls are drawn at random without replacement and all of them are found to be black, then the probability that the bag contains 1 red and 9 black balls is:
(A) $\frac{7}{110}$ (B) $\frac{7}{11}$
(C) $\frac{14}{55}$ (D) $\frac{7}{55}$
7. The value of $\lim_{x \rightarrow 0} \frac{\log_e(\sec(ex) \cdot \sec(e^2x) \cdot \dots \cdot \sec(e^{10}x))}{e^2 - e^{2 \cos x}}$ is equal to
(A) $\frac{(e^{10} - 1)}{2e^2(e^2 - 1)}$
(B) $\frac{(e^{20} - 1)}{2(e^2 - 1)}$
(C) $\frac{(e^{20} - 1)}{2e^2(e^2 - 1)}$
(D) $\frac{(e^{10} - 1)}{2(e^2 - 1)}$
8. For three unit vectors $\vec{a}, \vec{b}, \vec{c}$ satisfying $|\vec{a} - \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} - \vec{a}|^2 = 9$ and $|2\vec{a} + k\vec{b} + k\vec{c}| = 3$, the positive value of k is
(A) 3 (B) 4 (C) 6 (D) 5
9. If $\int \left(\frac{1-5\cos^2 x}{\sin^5 x \cos^2 x} \right) dx = f(x) + C$, where C is the constant of integration, then $f\left(\frac{\pi}{6}\right) - f\left(\frac{\pi}{4}\right)$ is equal to
(A) $\frac{2}{\sqrt{3}}(4 + \sqrt{6})$ (B) $\frac{1}{\sqrt{3}}(26 - \sqrt{3})$
(C) $\frac{1}{\sqrt{3}}(26 + \sqrt{3})$ (D) $\frac{4}{\sqrt{3}}(8 - \sqrt{6})$
10. Let A, B and C be three 2×2 matrices with real entries such that $B = (I + A)^{-1}$ and $A + C = I$. If $BC = \begin{bmatrix} 1 & -5 \\ -1 & 2 \end{bmatrix}$ and $CB \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 12 \\ -6 \end{bmatrix}$, then $x_1 + x_2$ is
(A) -2 (B) 0 (C) 4 (D) 2
11. The value of $\sum_{k=1}^{\infty} (-1)^{k+1} \left(\frac{k(k+1)}{k!} \right)$ is
(A) $\frac{2}{e}$ (B) $\sqrt{e}$
(C) $\frac{e}{2}$ (D) $\frac{1}{e}$
12. If $\frac{\tan(A-B)}{\tan A} + \frac{\sin^2 C}{\sin^2 A} = 1$, where $A, B, C \in (0, \frac{\pi}{2})$, then
(A) $\tan A, \tan C, \tan B$ are in G.P.
(B) $\tan A, \tan C, \tan B$ are in A.P.
(C) $\tan A, \tan B, \tan C$ are in A.P.
(D) $\tan A, \tan B, \tan C$ are in G.P.
13. The common difference of the A.P.: $a_1, a_2, \dots, a_m$ is 1 more than the common difference of the A.P. $b_1, b_2, \dots, b_n$. If $b_{31} = -277$, $b_{43} = -385$ and $a_{78} = 327$ then a_1 is equal to
(A) 16 (B) 19 (C) 24 (D) 21
14. Find the area of the region $R = \{(x, y) : xy \leq 8, 1 \leq y \leq x^2, x \geq 0\}$ is
(A) $\frac{2}{3}(20 \log_e(2) + 9)$
(B) $\frac{1}{3}(40 \log_e(2) + 27)$
(C) $\frac{2}{3}(24 \log_e(2) - 7)$
(D) $\frac{1}{3}(49 \log_e(2) - 15)$

15. Let $S = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$. Let x be the number of 9-digit numbers formed using the digits of the set S such that only one digit is repeated and it is repeated exactly twice. Let y be the number of 9-digit numbers formed using the digits of the set S such that only two digits are repeated and each of these is repeated exactly twice. Then,
- (A) $45x = 7y$ (B) $21x = 4y$
(C) $29x = 5y$ (D) $56x = 9y$
16. Let $y = x$ be the equation of a chord of the circle C_1 (in the closed half-plane $x \geq 0$) of diameter 10 passing through the origin. Let C_2 be another circle described on the given chord as its diameter. If the equation of the chord of the circle C_2 , which passes through the point $(2, 3)$ and is farthest from the center of C_2 , is $x + ay + b = 0$, then $a - b$ is equal to
- (A) -6 (B) -2 (C) 6 (D) 10
17. Let ABC be an equilateral triangle with orthocenter at the origin and the side BC on the line $x + 2\sqrt{2}y = 4$. If the co-ordinates of the vertex A are (α, β) , then the

greatest integer less than or equal to $|\alpha + \sqrt{2}\beta|$ is

- (A) 4 (B) 5 (C) 3 (D) 2

18. Let f be a polynomial function such that $f(x^2 + 1) = x^4 + 5x^2 + 2$, for all $x \in \mathbb{R}$. Then $\int_0^3 f(x) dx$ is equal to

- (A) $\frac{41}{3}$ (B) $\frac{27}{2}$
(C) $\frac{33}{2}$ (D) $\frac{5}{3}$

19. Let $y = y(x)$ be the solution of the differential equation $x \frac{dy}{dx} - \sin 2y = x^3 (2 - x^3) \cos^2 y$, $x \neq 0$. If $y(2) = 0$, then $\tan(y(1))$ is equal to

- (A) $\frac{3}{4}$ (B) $\frac{7}{4}$
(C) $-\frac{3}{4}$ (D) $-\frac{7}{4}$

20. If the distances of the point $(1, 2, a)$ from the line $\frac{x-1}{1} = \frac{y-2}{1} = \frac{z-a}{1}$ along the lines $L_1 : \frac{x-1}{3} = \frac{y-2}{4} = \frac{z-a}{b}$ and $L_2 : \frac{x-1}{1} = \frac{y-2}{4} = \frac{z-a}{c}$ are equal, then $a + b + c$ is equal to

- (A) 7 (B) 6 (C) 4 (D) 5

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21. In a G.P., if the product of the first three terms is 27 and the set of all possible values for the sum of its first three terms is $\mathbb{R} - (a, b)$, then $a^2 + b^2$ is equal to ____.
22. For some $\theta \in (0, \frac{\pi}{2})$, let the eccentricity and the length of the latus rectum of the hyperbola $x^2 - y^2 \sec^2 \theta = 8$ be e_1 and l_1 , respectively, and let the eccentricity and the length of the latus rectum of the ellipse $x^2 \sec^2 \theta + y^2 = 6$ be e_2 and l_2 , respectively. If $e_1^2 = e_2^2 (\sec^2 \theta + 1)$, then $(\frac{l_1 l_2}{e_1 e_2}) \tan^2 \theta$ is equal to ____.

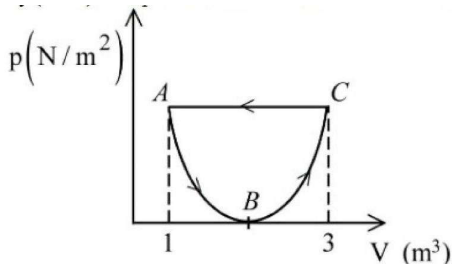
23. If $k = \tan(\frac{\pi}{4} + \frac{1}{2} \cos^{-1}(\frac{2}{3})) + \tan(\frac{1}{2} \sin^{-1}(\frac{2}{3}))$, the number of solutions of the equation $\sin^{-1}(kx - 1) = \sin^{-1} x - \cos^{-1} x$ is ____.

24. The value of $\sum_{r=1}^{20} \left(\left| \sqrt{\pi \left(\int_0^r x |\sin \pi x| dx \right)} \right| \right)$ is ____.

25. Let PQR be a triangle such that $\vec{PQ} = -2\hat{i} - \hat{j} + 2\hat{k}$ and $\vec{PR} = \hat{a}\hat{i} + b\hat{j} - 4\hat{k}$, $a, b \in \mathbb{Z}$. Let S be the point on QR which is equidistant from the lines PQ and PR . If $|PR| = 9$ and $\vec{PS} = \hat{i} - 7\hat{j} + 2\hat{k}$, then the value of $3a - 4b$ is ____.

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26. In the following p - V diagram the equation of state along the curved path is given by $(V - 2)^2 = 4ap$ where a is a constant. The total work done in the closed path is



- (A) $\frac{1}{2a}$ (B) $-\frac{1}{3a}$
(C) $+\frac{1}{3a}$ (D) $-\frac{1}{a}$

27. Which of the following best represents the temperature versus heat supplied graph for water, in the range -20°C to 120°C ?

